

Assignment 1

a. Using the confusion matrix, manually calculate the Accuracy value. Does it match the value calculated by your program? If not, why? (Manually includes using a spreadsheet).

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Samples}}$$

$$\text{Accuracy} = \frac{50 + 48 + 47}{150} = \frac{145}{150} = 0.967$$

It matches the program output.

b. Using the confusion matrix, manually calculate the Precision values for each iris variety. Do they match the values calculated by your program? If not, why?

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$$

$$\text{Precision}_{\text{Iris-Setosa}} = \frac{50}{50} = 1$$

$$\text{Precision}_{\text{Iris-Versicolor}} = \frac{48}{51} = 0.94$$

$$\text{Precision}_{\text{Iris-Virginica}} = \frac{47}{49} = 0.96$$

It matches the program output.

c. Using the confusion matrix, manually calculate the Recall values for each iris variety. Do they match the values calculated by your program? If not, why?

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}}$$

$$\text{Recall}_{\text{Iris-Setosa}} = \frac{50}{50} = 1$$

$$Recall_{Iris-Versicolor} = \frac{48}{50} = 0.96$$

$$Recall_{Iris-Virginica} = \frac{47}{50} = 0.94$$

It matches the program output.

d. Using the confusion matrix, manually calculate the F1 values for each iris variety. Do they match the values calculated by your program? If not, why?

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

$$F1_{Iris-Setosa} = \frac{2 \times 1 \times 1}{1 + 1} = 1$$

$$F1_{Iris-Versicolor} = \frac{2 \times 0.94 \times 0.96}{0.94 + 0.96} = 0.95$$

$$F1_{Iris-Virginica} = \frac{2 \times 0.96 \times 0.94}{0.96 + 0.94} = 0.95$$

It matches the program output.

Output from CheckVersions.py:

```
python: 3.11.4 (v3.11.4:d2340ef257, Jun 6 2023, 19:15:51) [Clang 13.0.0 (clang-1300.0.29.30)]
scipy: 1.16.1
numpy: 2.2.3
pandas: 2.3.2
sklearn: 1.7.1
hello world!
```

Output from NBClassifier.py

```
Gaussian Naive Bayes Classifier (2-Fold CV)
```

```
Accuracy: 0.967
```

```
Confusion Matrix:
```

```
[[50  0  0]
```

```
 [ 0 48  2]
```

```
 [ 0  3 47]]
```

```
Classification Report:
```

	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	50
versicolor	0.94	0.96	0.95	50
virginica	0.96	0.94	0.95	50
accuracy			0.97	150
macro avg	0.97	0.97	0.97	150
weighted avg	0.97	0.97	0.97	150